

DESIGN AND ANALYSIS OF CUTTING BLADE FOR ROTARY TYPE GRASS CUTTERS

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ABSTRACT

This paper presents the design and analysis of a blade for rotary type grass cutters. The study investigates the addition of triangular fins to the blade in order to improve efficiency and reduce the drag force. The blade was analyzed in the ANSYS Fluent software to simulate the air flow around it. It has been shown by the results that the addition of triangular fins reduced turbulence and improved the efficiency of the blade. The drag force was reduced by a high percentage compared to the blade without fins. The blade was also analyzed for various forces acting on it to check the structural integrity of the model. The results showed that the blade was capable of withstanding the various forces that it would experience during operation. Overall, this study provides important insights into the design of blades for rotary type grass cutters, and highlights the importance of considering aerodynamics in blade design to improve efficiency and reduce drag force.

Keywords: Rotary Lawn Mower, Cutting Blade, Blade Design, CFD, Static Structural Analysis, Cutting Performance.

I. INTRODUCTION

A rotary type grass cutter is an essential tool for maintaining lawns, gardens, and parks. It is a versatile device that uses a rotating blade to cut grass and other vegetation. The design and efficiency of the blade are critical for the performance of the cutter. An optimized blade design can reduce the time and effort required for lawn maintenance while also increasing the longevity of the cutter. Previous studies have shown that the deformation, stress, and strain energy of the blade are crucial parameters that determine the structural integrity and durability of the blade. Therefore, the analysis of the deformation, stress, and strain energy in the blades of a rotary type grass cutter is a necessary step in understanding the cutting performance and durability of the blade. Therefore, there is a need for research to develop and analyze the blade design for rotary type grass cutters.

The best blade cutting angle for forage chopping, is normally between 30 and 40 degrees, with an ideal cutting-edge radius of about 0.05 mm. The minimal power needed for rotary mowers is attained with blade angles between 25 and 30 degrees. A high impact velocity is required, as demonstrated by field tests on mowers and laboratory tests on single stems. Mowers with a disc or a rotary blade generally run between 71 and 84 m/s. [1] Xu et al. [2] developed a new type of silage knife by using laser cladding technology to reinforce a WC-high entropy alloy coating onto the blade surface. The authors evaluated the performance of the new silage knife, including cutting efficiency, wear resistance, and self-grinding ability. The results showed that the WC-high entropy alloy reinforced silage knife had a longer lifespan and improved cutting efficiency compared to conventional silage knives. The self-grinding ability of the new silage knife was also found to be improved, which can further reduce energy consumption and maintenance costs. S Choi et al. [3] analyzed the deformation behavior of a cutting blade for a rotary type grass cutter using finite element analysis. The authors evaluated the blade's stress distribution, deformation behavior, and cutting force, and the results showed that the blade's deformation behavior can significantly affect its cutting performance and durability. K. Hwang and Y. Kim [4] used the Taguchi method to optimize the blade design parameters, such as blade shape, material, and coating, for a rotary type grass cutter. The authors evaluated the blade's cutting performance and durability using experimental tests and statistical analysis, and the results showed that the optimized blade design can significantly improve the cutting performance and durability of the grass cutter. The maximum deformation of the blade occurs at the cutting edge, which is affected by the cutting angle, cutting depth, and the shape of the cutting edge. The maximum stress occurs at the root of the blade, which is affected by the shape and thickness of the blade. The strain energy density is concentrated near the cutting edge and the root of the blade. [5] Other factors such as angle of attack significantly affected the air flow and lift force, which can affect the cutting

efficiency and fuel consumption [6]. Increasing the blade angle can improve the blade's lift-to-drag ratio and reduce the drag force [7].

The aim of this research paper is to investigate the design and analysis of cutting blades for rotary type grass cutters. In this paper, the variation of blade shape has been analyzed using Finite Element Method, to evaluate total deformation, stress distribution and the cutting force of the blade. To improve the efficiency of the blade and reduce the drag force a triangular fin has been added and analysis have been done using CFD. The results of the study will provide a deeper understanding of the design and performance of cutting blades for rotary type grass cutters for a more efficient and effective grass cutting process.

II. DEVELOPMENT OF MODEL

The design of the cutting blade is based on existing designs, with few modifications to increase the efficiency. For the analysis the cutting angle of the blade has been taken as 20° and the cutting-edge radius as 0.001 mm, as shown in Fig 1.

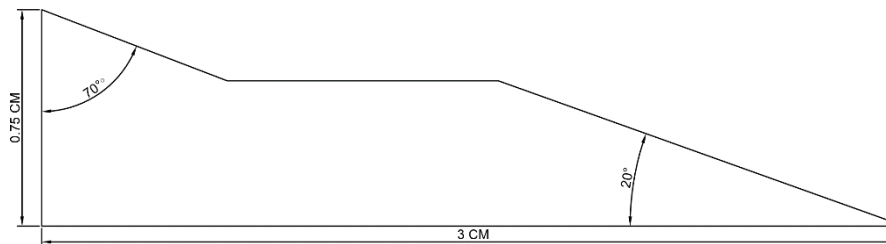


Fig 1: Blade cutting angle of 20° (Side profile)

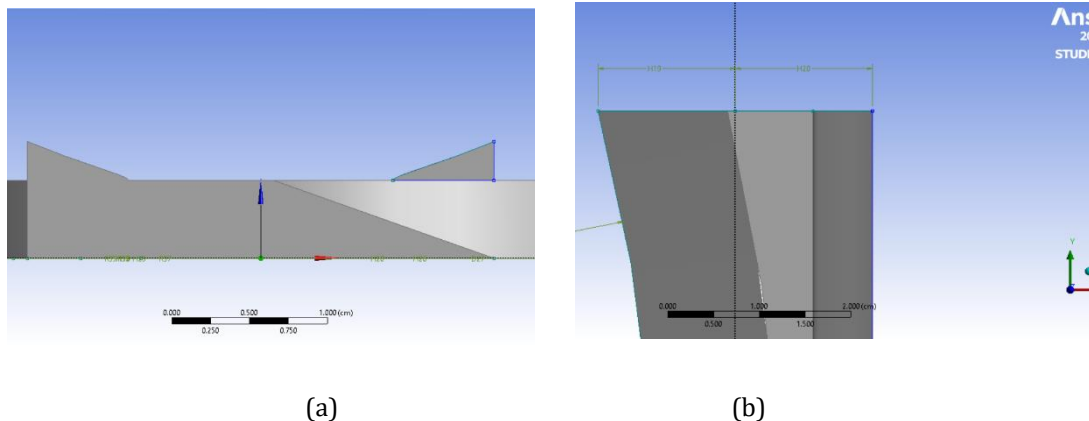


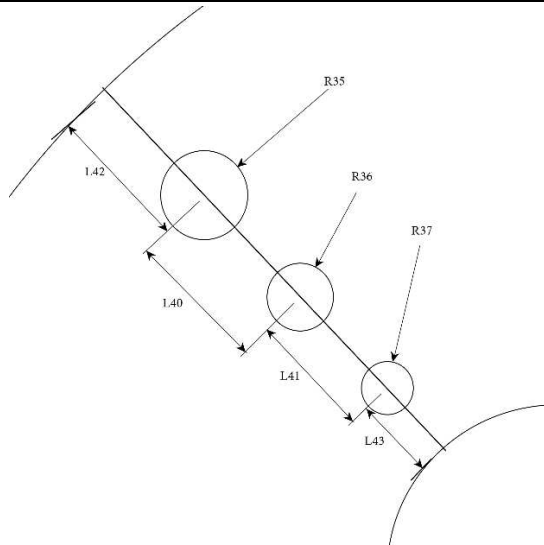
Fig 2: Different view of the blade model (a) Side View (b) Front View

The use of stainless steel as a blade material improves the blade's corrosion resistance and durability, making it a suitable material for grass cutter blades [8]. The grass found in the region are flood – plain grass, which are not harsh for grass cutting applications, and thus Stainless Steel is selected as the blade material:

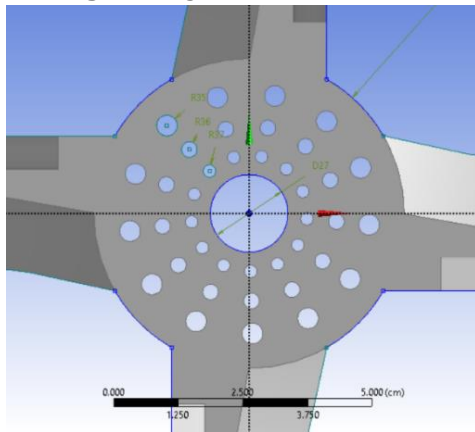
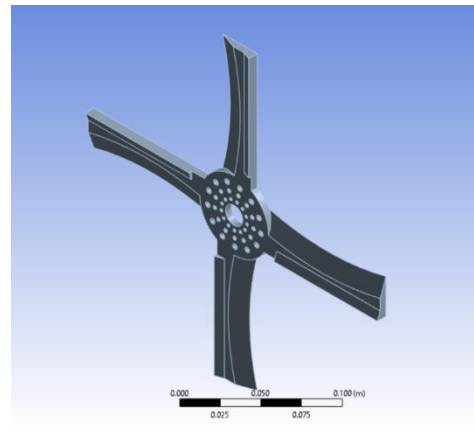
Table 1: Material Property for Stainless Steel selected for blade design [9]

Density	7750 Kg/m ³
Youngs Modulus	1.93E+11 Pa
Shear Modulus	7.3664E+10 Pa
Poisson's Ratio	0.31

Ventilation/Cooling holes, shown in Fig 4, were added in the central disk of the blade because of two purposes. Firstly, to reduce the weight of the solid blade. Secondly, in case of application of emergency brakes when the rotor is at full capacity, these will improve the cooling performance of the overall blade and hence reducing the peak high temperature reached. The design of the ventilation hole, i.e., shown in Fig 3, is based on the study of H. Li et al. [10]. The design parameters assumed from the reference, for the development of the ventilation hole is shown in Table 2, 'R' refers to the radius of the ventilation holes and 'L' refers to the distance between different edges, which is shown in Fig 3.


Fig 3: Design of Ventilation Holes

Dimensions	Value
R35	0.2 Cm
R36	0.15 Cm
R37	0.12 Cm
L41	0.6 Cm
L40	0.6 Cm
L42	0.6 Cm
L43	0.3 Cm

Table 2: Dimension of Ventilation Holes

Fig 4: Ventilation Holes on Model

Fig 5: Isometric View of the model

III. MODEL PROPERTIES AND CALCULATIONS

Model Properties from ANSYS

For the analysis of the model the following data from mass inertia properties of ANSYS workbench has been used.

Mass =	0.396 Kg
Volume =	$5.1197 \times 10^{-5} \text{ m}^3$
Surface Area=	$3.8 \times 10^{-3} \text{ m}^2$
Moment of Inertia (about Z axis) =	$2.2239 \times 10^{-3} \text{ Kg m}^2$

Calculation of shearing force

$$\text{Shearing Force } (\sigma) = \frac{\tau}{r \times \sin \delta} \text{ N}$$

$$\alpha = \frac{2 \times \pi \times \omega}{60} = \frac{2 \times \pi \times 2500}{60} = 261.8 \text{ rad/S}^2$$

$$\tau = I \times \alpha = 0.0022239 \times 261.8 = 0.5822 \text{ N.m}$$

$$\sigma = \frac{0.5822}{0.001 \times \sin 20^\circ} = 1702.23 \text{ N}$$

The Shearing force for the blade at maximum RPM of 2500 is calculated to be 1702.23 N, similarly shearing force for rotations speeds of 500, 1000, 1500, 2000 RPM is calculated and plotted. The resultant graph, as shown in Fig 6, show a linear variation of shearing force with increase in RPM.

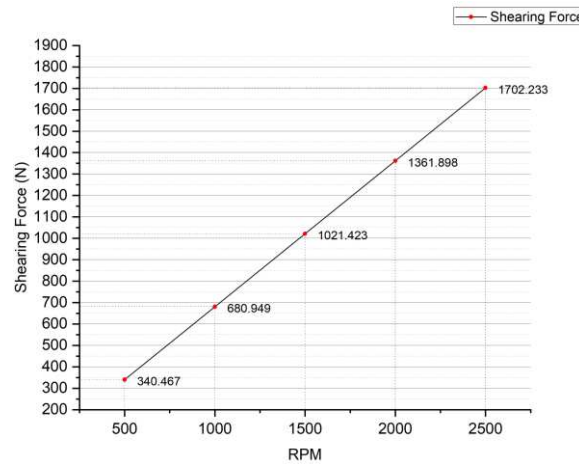


Fig 6: RPM vs Shearing Force

Calculation of Drag Force from Drag Coefficient

$$D = \left(\frac{1}{2}\right) \times \rho \times V^2 \times c_d \times A \text{ N}$$

$$D = \left(\frac{1}{2}\right) \times 7750 \times 45.7^2 \times 0.010062 \times 0.0313721033 \text{ N}$$

$$D = 7.07 \text{ N}$$

IV. RESULTS AND DISCUSSION

Aerodynamic analysis of the blade

Due to the blade's high-speed rotation, an effect known as the vortex effect occurs at the trailing edge of the blade. The vortex effect makes the back side of the blade's turbulence worse, decreasing the blade's effectiveness. Several designs were tried in an effort to prevent this by reducing turbulence, and the addition of a Triangular Fin to the back end was found to be the most successful. This also decreased the drag and downforce of the blade and increased the efficiency of cutting. A with the dimensions 59.06 x 34.33 x 100 mm was made to evaluate the blade's aerodynamic behavior. FLUENT Analysis showed that the turbulence at the rear end of the blade reduced significantly after addition of Triangular Aero foil, also resulting in the reduction of drag force by 7.12% giving a value of 7.07 N. Previous works in different blade designs resulted in similar improvements in the drag force [11]. The velocity vector is shown in Fig 7 and the total pressure contours are shown in Fig 8.

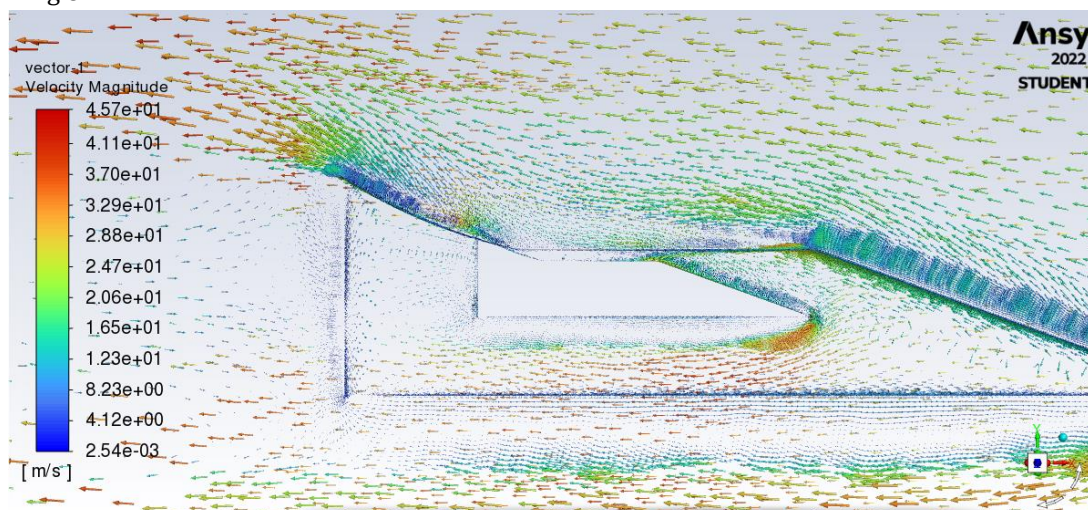


Fig 7: Velocity Vector showing reduced turbulence at the rear end of the blade

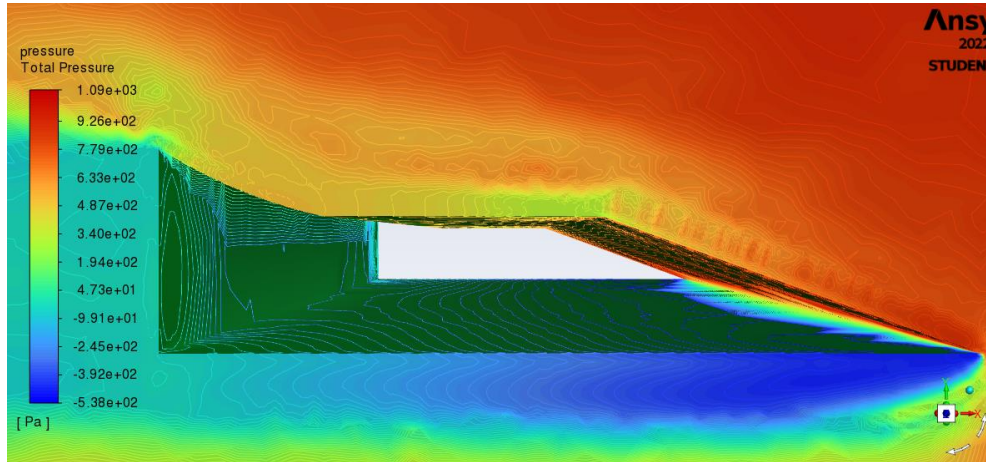


Fig 8: Contours of total pressure in the fluid domain

Static structural analysis of the blade

For the Static Structural Analysis of the blade the mounting bracket is taken as fixed support. Drag is taken as pressure force, acting in the Y-Z plane and the shearing force is taken tangentially to the plane of rotation. The blade is meshed finely, which is shown in Fig 32, with face sizing refinement provided to the cutting edge and fin portion for more accurate results. The resulting mesh contained 100875 Nodes and 54724 elements. The element order is selected to be quadratic, since the blade model consists of many curves and edges, it provides a smooth and continuous cutting profile. Also, the quadratic meshing creates a more aerodynamic profile.

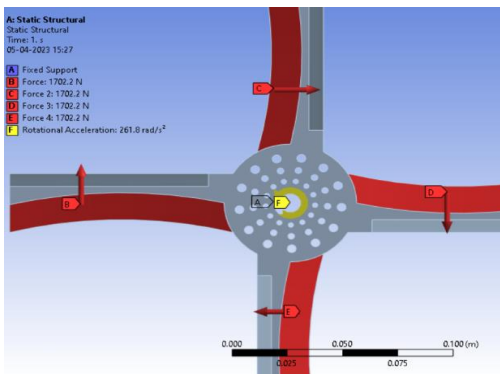


Fig 9: Refined Mesh

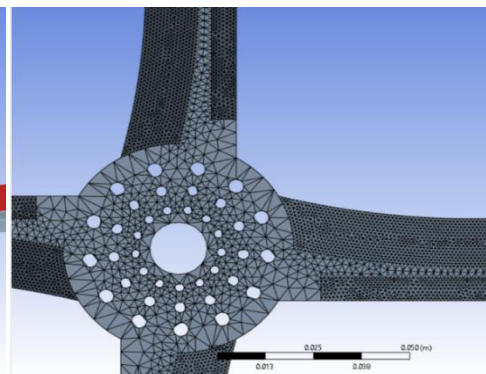


Fig 10: Boundary Conditions

Table 3: Static Structural Analysis of Model at 2500 RPM [8]

Parameter	Maximum
Deformation	$9.6487 \times 10^{-4} \text{ m}$
Strain Energy	$3.4227 \times 10^{-3} \text{ J}$
Normal Stress	$6.476 \times 10^8 \text{ Pa}$
Shear Stress	$2.7857 \times 10^8 \text{ Pa}$
Von mises stress	$6.7049 \times 10^8 \text{ Pa}$
Von mises strain	3.49×10^{-3}

After the static structural analysis of the blade model at 2500 RPM, the results of the total deformation are shown in Fig 11, Strain Energy is shown in Fig 12, Normal Stress is shown in Fig 13, Shear Stress is shown in Fig 14.

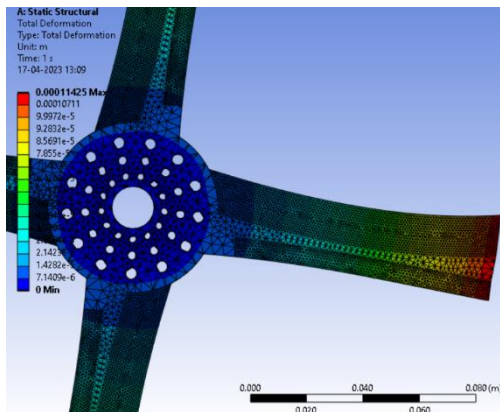


Fig 11: Total Deformation

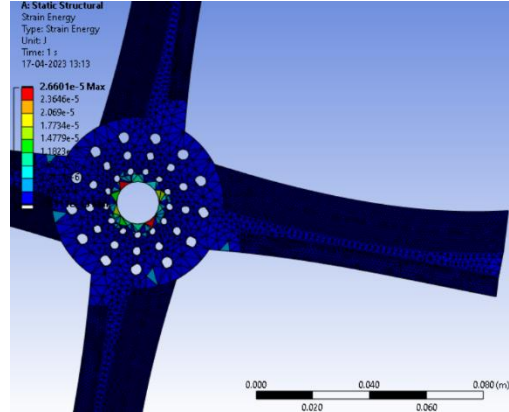


Fig 12: Strain Energy distribution

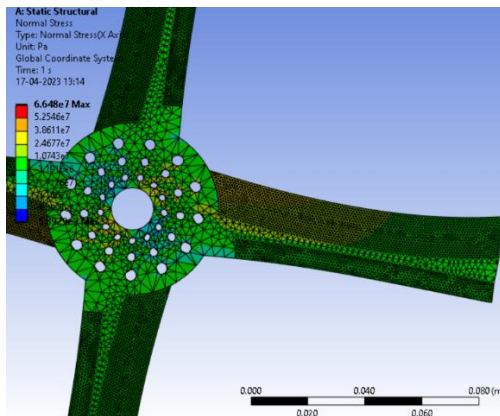


Fig 13: Normal Stress distribution

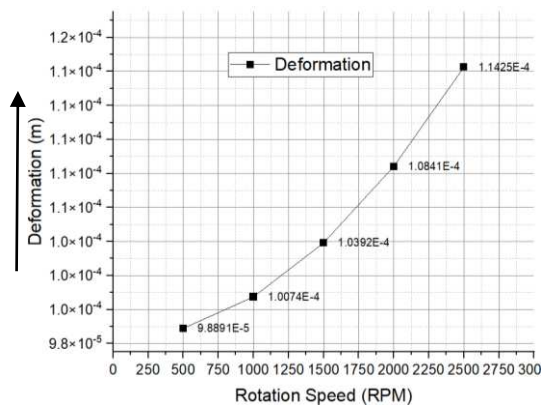
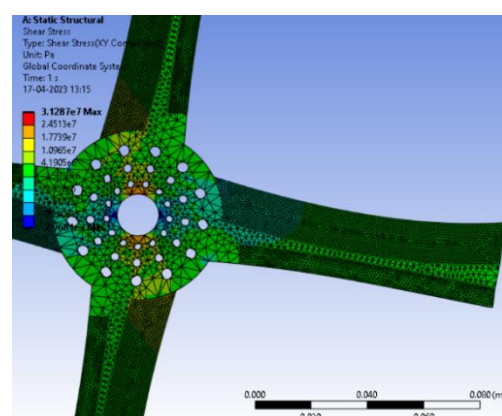


Fig 15: Total deformation vs blade speed (RPM)

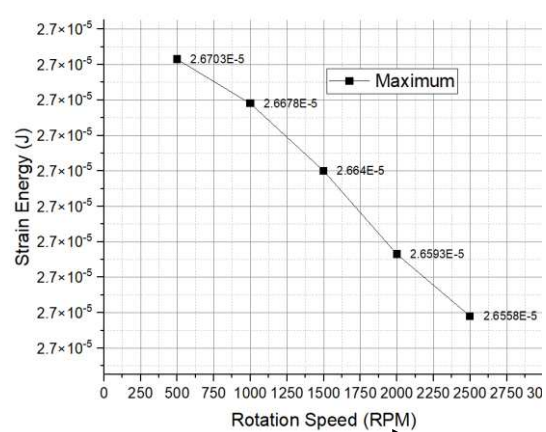


Fig 16: Strain energy vs blade speed (RPM)

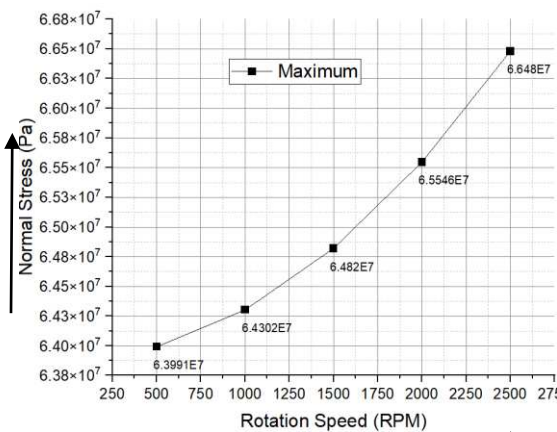


Fig 17: Normal Stress vs blade speed (RPM)

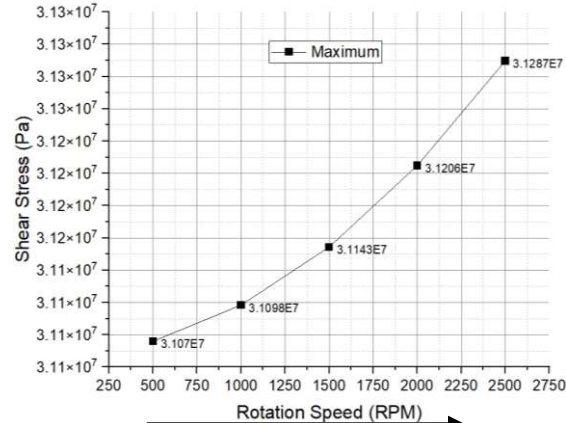


Fig 18: Shear Stress vs blade speed (RPM)

Fig 15, 16, 17, 18 shows the variation of Total Deformation, Strain Energy, Normal Stress and Shear Stress with rotational velocity (RPM) of the blade. The equivalent stress (von-mises stress) is less than the yield strength of the material, thus the material is loaded within its elastic limit [9]. The reduction in strain energy with increase in rotation speed is owing to the fact the blade produces lesser vibration at higher speed. Results were compared with previous works [11], and it was concluded that the changes in blade designs resulted in better blade structural integrity at higher blade speed (RPM).

V. CONCLUSION

The aim of developing an efficient rotary lawn cutting blade design has been successfully achieved. FLUENT analysis of the current basic blade design created a lot of undesirable turbulence resulting in inadequate grass cutting and reduction in efficiency of the process. These restrictions were substantially removed in the final layout. The blade's design was improved through a fluid analysis that produced acceptable outcomes. The new blade design was also discovered to be secure in the face of various forces while turning. The layout has further scope for development by study and testing. The validation of results from previous works [11] showed significant improvement in the aerodynamic performance of a four-blade rotary lawn mower cutting blade, with increased structural integrity and properties.

VI. REFERENCES

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